Manniesoma Lab Project

Cybersecurity Infrastructure and DevSecOps Portfolio

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Project record | October 2026

Project Overview

Designed and documented a virtual cybersecurity lab with Proxmox, OPNsense, Active Directory, DNS, Linux, and a Windows domain client. Demonstrated domain authentication, standard-user access, and logged IPv4 network isolation through reproducible checks and supporting screenshots.

Demonstrated Capabilities

- Infrastructure operations: provisioned virtual machines, configured network services, and diagnosed connectivity issues.
- Security validation: checked identity, permissions, DNS discovery, and firewall enforcement against observable results.
- Technical documentation: preserved configuration values, troubleshooting steps, evidence, and acceptance criteria.

Career Direction

DevSecOps Engineer, with a focus on translating infrastructure and security fundamentals into version-controlled configuration, automated testing, and secure software delivery.

Credentials: CompTIA A+ and Network+ earned. Security+ planned for January 2027.

Project Status

The lab is functional through domain sign-in and least-privilege validation. The Step 4 workstation OU move is documented as the next controlled change; its completion remains to be verified. Future CI/CD and cloud projects are identified separately from completed work.

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1 Goal

I built a virtual cybersecurity lab to practice network security, Linux administration, Windows identity management, and troubleshooting on systems under my control. The functional milestone is a Windows client that can find DC1 through DNS, authenticate to cyberlab.test, and operate with a standard domain account behind OPNsense.

This foundation supports my ultimate goal of becoming a DevSecOps Engineer. The next stage is to turn documented manual work into version-controlled configuration, automated validation, and secure application delivery.

Milestone	Recorded result
Virtual infrastructure	Proxmox pve01 and Ubuntu, firewall, Windows Server, Kali, and Windows client VMs
Network boundary	OPNsense separates the 10.10.10.0/24 lab from the 192.168.50.0/24 home network; logged IPv4 blocks demonstrate enforcement
Identity and DNS	DC1 at 10.10.10.10 serves cyberlab.test; domain login and group membership verified
Workstation organization	Workstations OU created and WIN11-CLIENT01 located; Step 4 is the move into that OU

2 Legal and Ethical Scope

All testing is limited to my own devices, my own network, intentionally vulnerable lab machines, and explicitly authorized training platforms. This project does not authorize testing public systems or other people's devices. Any future vulnerable application or security-testing target belongs in a dedicated isolated segment.

The build record ends at Step 4 of the final setup post in Continue Lab Configuration. The completed evidence establishes the state immediately before the move; the move is documented as the next controlled change. Later GPO creation and hardening are future work.

3 Required Tools

Resource	Role in this project
Proxmox VE on pve01	Hosts and manages the virtual lab; October screenshots display VE 9.2.21
OPNsense FW01 VM 101	Routes and filters lab traffic; console displays OPNsense 26.7
Ubuntu test VM 100	Linux services, SSH, guest-agent troubleshooting, and future automation
Windows Server DC1 VM 102	Active Directory Domain Services and DNS
Kali Linux VM 103	Authorized internal validation and firewall testing
Windows client VM 104	WIN11-CLIENT01 domain sign-in and standard-user validation
PowerShell and Command Prompt	Windows network, identity, directory, and policy checks
Browser and Proxmox console	VM installation, console recovery, and firewall management
Screenshots and PDF binder	Build history, troubleshooting evidence, and repeatable documentation

Personal workstation inventory

My Yoga 9i has 32 GB RAM, a 1 TB SSD, an Intel Core Ultra 7 processor, and a 4K 60 Hz display. My Area-51m R2 has an Intel i9, 32 GB RAM, 4 TB SSD storage including 2 TB NVMe, and a 360 Hz display. Both connect to the ASUS XG27UCS monitor using the documented docking and display connections.

These workstations support administration, development, and lab access.

Career context

I have A+ and Network+ and plan to take Security+ by January 2027. My stated academic plan is to complete the cybersecurity associate degree in Fall 2026, continue engineering transfer preparation during 2027–2028, and target CSUN in Spring 2029. These are personal goals, not completed credentials or guaranteed transfer milestones.

4 Network Safety Notes

Recorded logical topology

```
Administration workstation
  ↓
Home network 192.168.50.0/24
  ├ Proxmox pve01: 192.168.50.222 (HTTPS port 8006)
  └ OPNsense FW01 WAN: 192.168.50.83
      ↓
OPNsense LAN: 10.10.10.1/24
      ↓
Lab network 10.10.10.0/24
  • DC1: 10.10.10.10 — Active Directory and DNS
  • WIN11-CLIENT01: 10.10.10.108 — domain client
  • Kali: 10.10.10.131 — internal validation
Domain: cyberlab.test

Ubuntu's earlier address was 192.168.50.213.
Verify its current interface and subnet before automation.
```

Addresses are recorded observations. DHCP addresses can change. The firewall WAN address is on the home network; the diagram is logical and does not imply that lab VMs have a direct household-network adapter.

Controls and limits

The BLOCK_LAB_TO_HOME rule and live log entries show lab-originated IPv4 traffic being blocked toward home addresses. Keep the deny rule ahead of broad allow rules and verify applied changes with logs. A failed ping alone is not enough evidence of isolation.

An IPv6 allow rule is visible in an intermediate rules screenshot. IPv4 blocking does not prove IPv6 isolation. Before adding vulnerable workloads, validate the IPv6 policy, management access, bridge membership, and any alternate adapters. Devices on the same lab subnet can communicate without crossing OPNsense, so later vulnerable workloads need their own segment or host controls.

Keep administrative interfaces private. Record configuration backups and recovery access before network changes. A VM snapshot can support a rollback but does not replace an independent backup and restore test.

5 Step by Step Setup

Stage A Establish the virtualization and Linux foundation

The early binder records Proxmox installation media preparation, repository screens, Ubuntu installation, network checks, and guest-agent troubleshooting. Ubuntu provided the first working Linux administration target before the later segmented Windows lab.

On the earlier Ubuntu system, I checked the hostname, IP addresses, SSH service, and QEMU guest agent. The final check shows both services active. Earlier “static unit” messages were part of troubleshooting; the useful outcome was the service state, not whether systemctl enable printed a success message.

The Ubuntu address 192.168.50.213 belongs to this earlier phase and must not be reused as the current isolated-lab address without checking it.

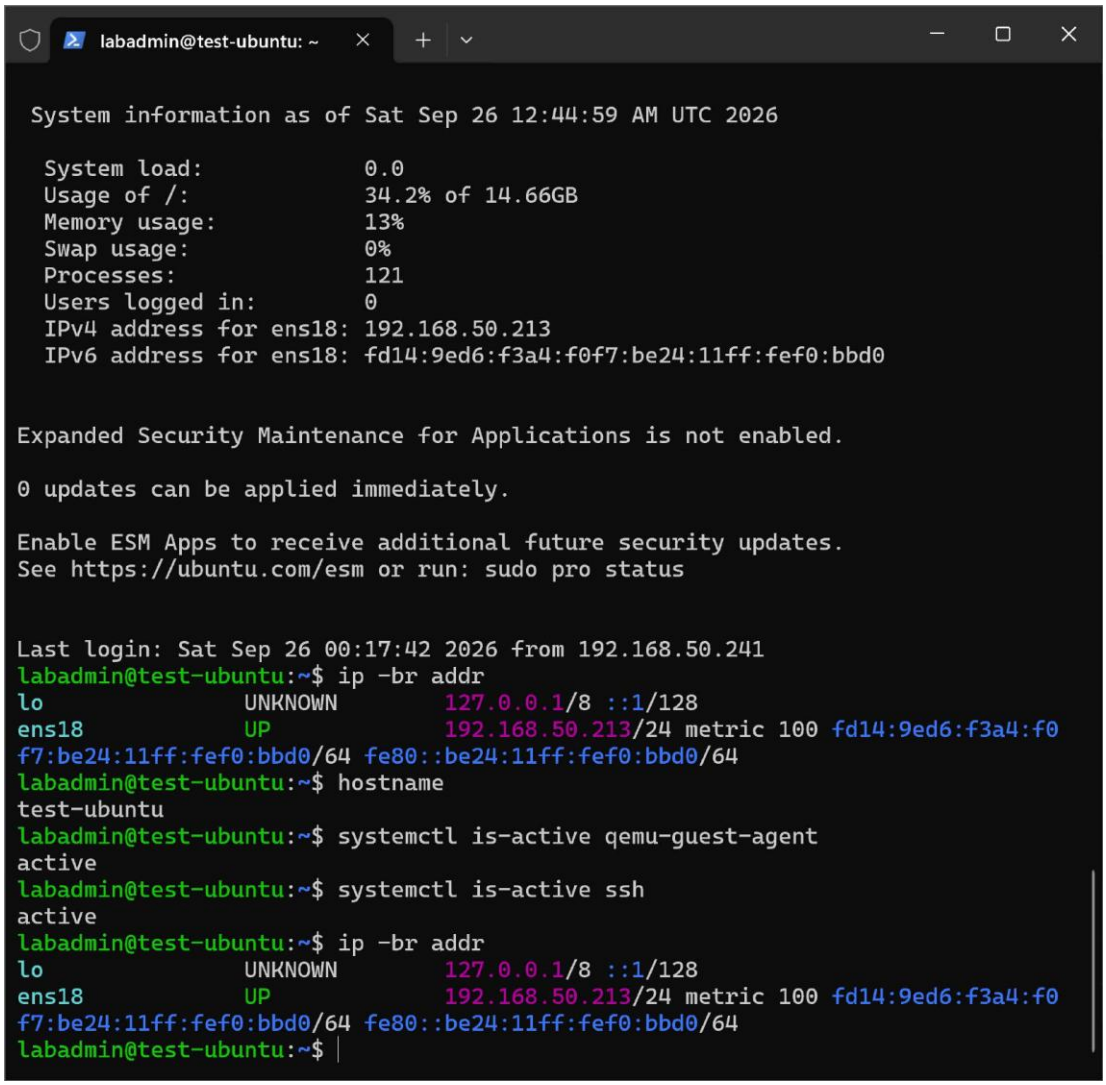


Figure 1. Earlier Ubuntu milestone: test-ubuntu, interface address, and active SSH and QEMU guest agent. Source: Binder1.pdf, page 25.

DevSecOps relevance: inventory and service checks become the first automated readiness tests. A configuration script should report both its intended change and a post-change verification result.

Firewall provisioning and addressing

I created FW01 as VM 101 and assigned separate WAN and LAN interfaces. The LAN used 10.10.10.1/24; the WAN later received 192.168.50.83/24. The early console setup specified a DHCP pool of 10.10.10.100–10.10.10.199 and retained HTTPS for the web interface.

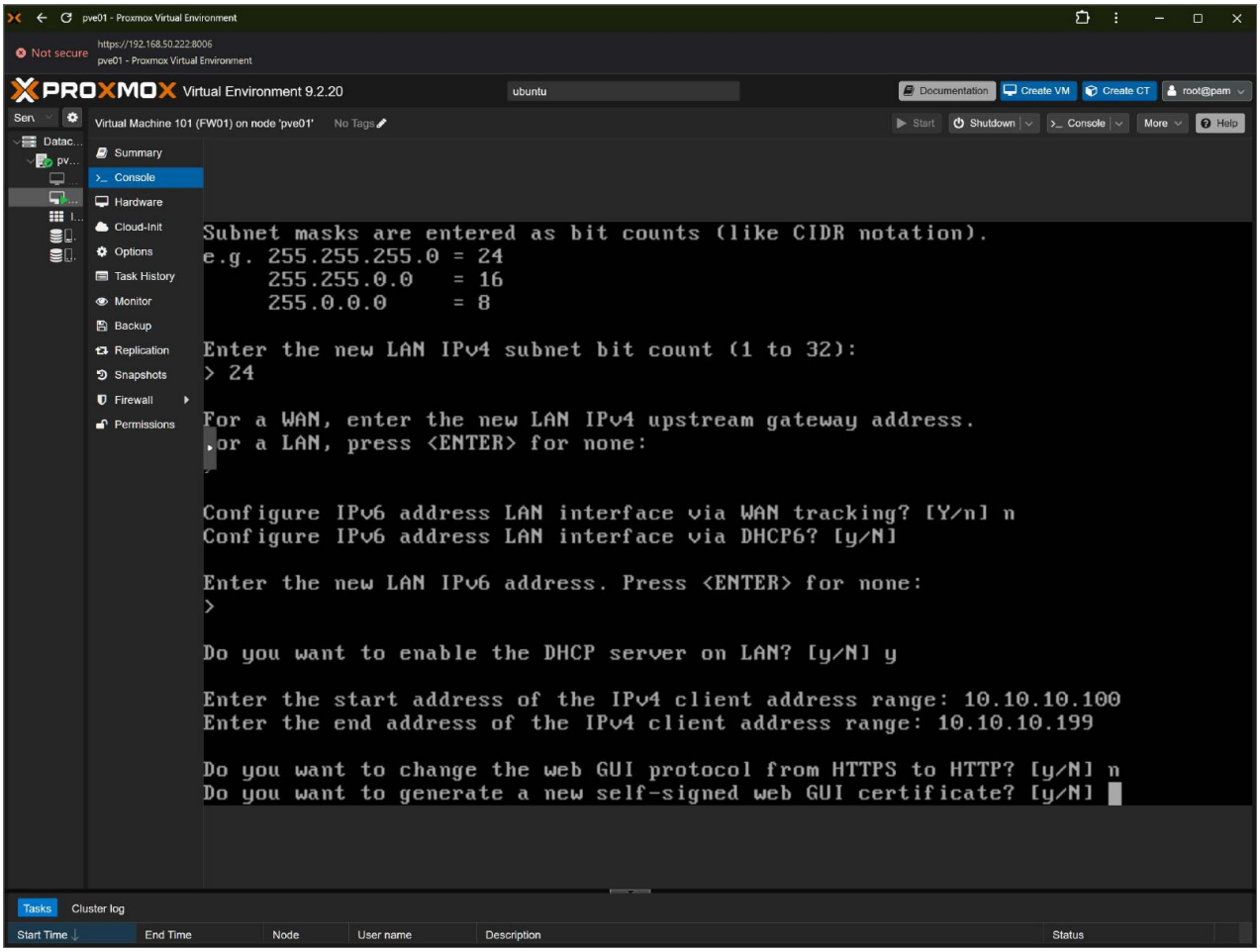


Figure 2. Initial LAN setup includes /24 addressing and DHCP pool 10.10.10.100–199. Source: Binder1.pdf, page 37.

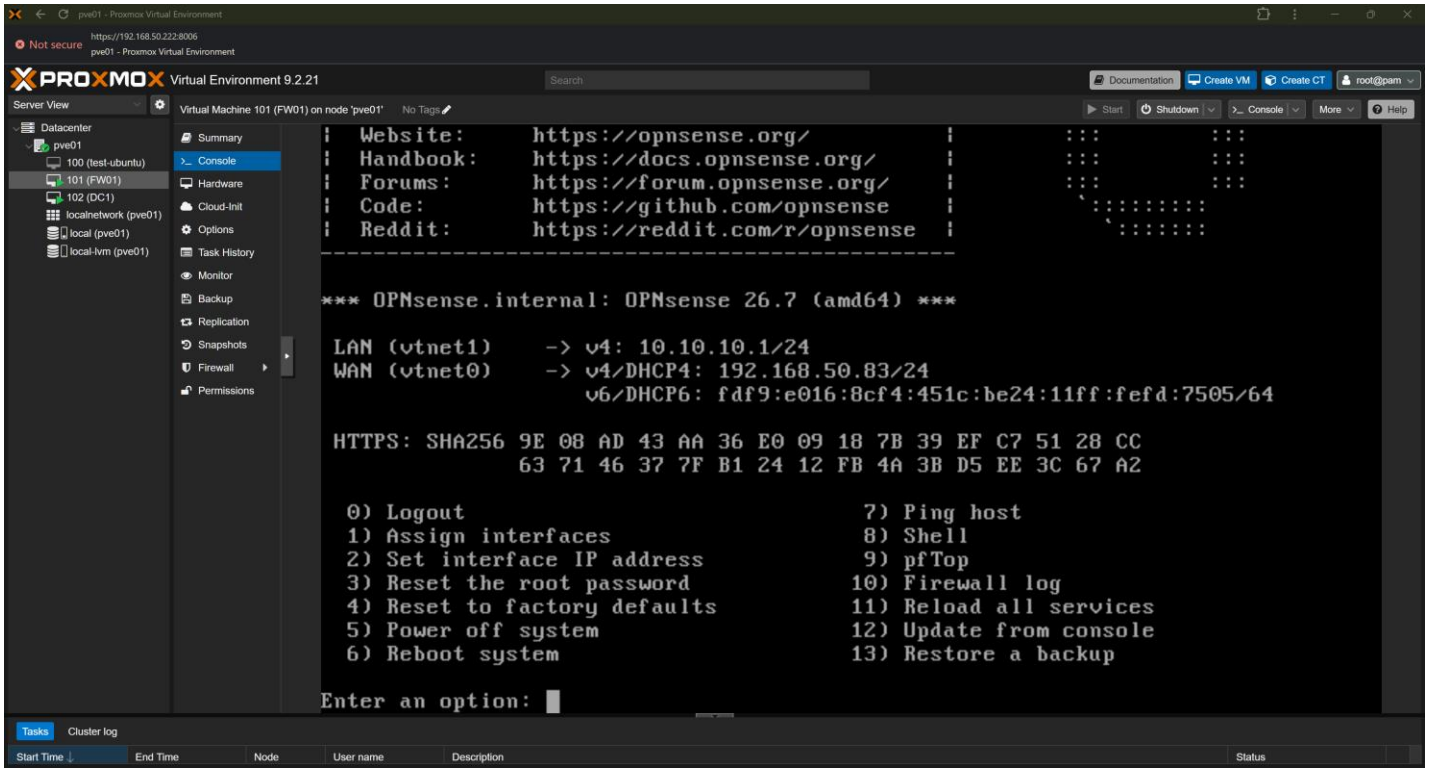


Figure 3. FW01 console confirms LAN vtnet1 at 10.10.10.1 and WAN vtnet0 at 192.168.50.83.

Later Dnsmasq screens show the service enabled while Kea is disabled. The original pool is historical configuration evidence; confirm the active service's current ranges and lease options before changing DHCP.

Firewall isolation and validation

I installed Kali and used it inside the lab to test connectivity and inspect OPNsense. I configured a lab-to-home block rule and checked the live firewall log to determine whether the intended traffic matched that rule.

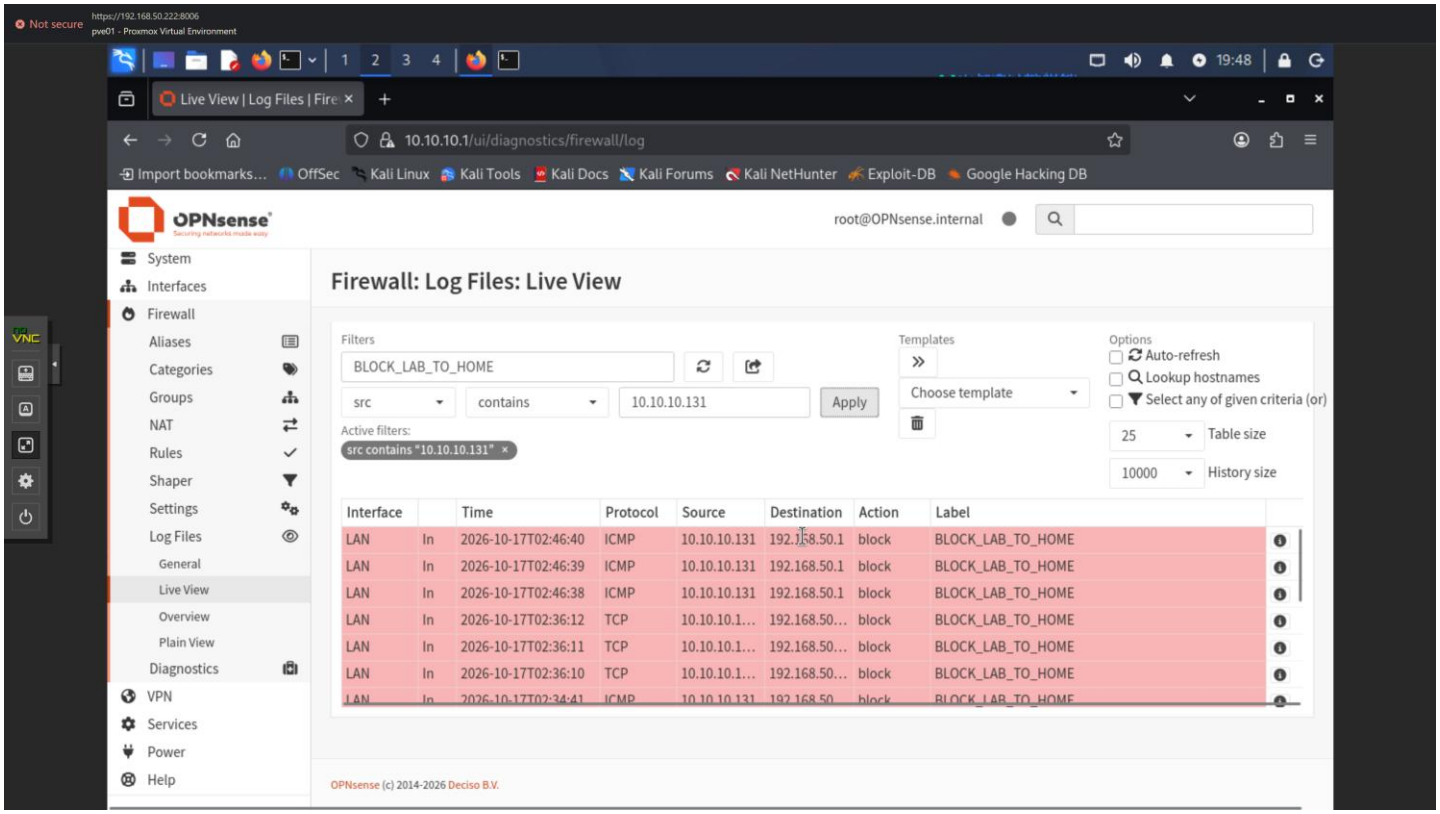


Figure 4. OPNsense logs BLOCK_LAB_TO_HOME drops from Kali at 10.10.10.131 toward 192.168.50.1.

The useful proof is the combination of source, destination, action, and rule label. Earlier rule-edit screenshots still show an Apply prompt, so they describe an intermediate configuration rather than sufficient proof that enforcement was active.

How the lab became usable without removing the boundary

I accessed management through the intended network paths and used VM consoles when remote access failed. Traffic tests distinguished gateway reachability, internet connectivity, and blocked access toward home systems. I did not need to place every lab machine directly on the home network.

DevSecOps connection

This is an early example of a security acceptance test: required traffic works, prohibited traffic is denied, and the result is observable. Future deployment checks should test both allowed and denied paths and retain logs as build evidence.

Domain controller and DNS foundation

Windows Server was installed as DC1. I configured a static lab address of 10.10.10.10/24 with gateway 10.10.10.1, installed the directory and DNS roles, and established cyberlab.test. The server's DNS configuration points to itself, including the IPv6 loopback entry shown in the capture.

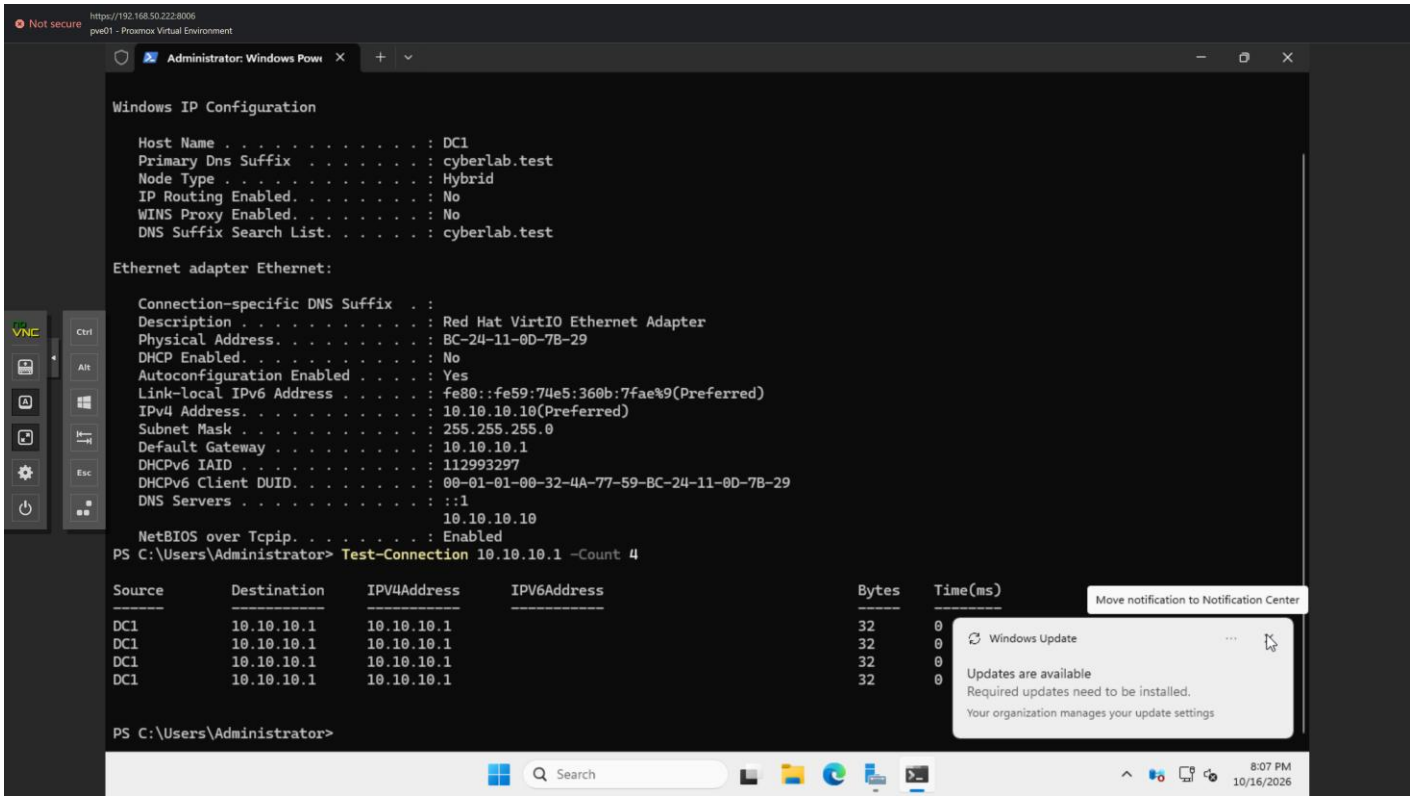


Figure 5. DC1 has static address 10.10.10.10/24 and successfully reaches its lab gateway.

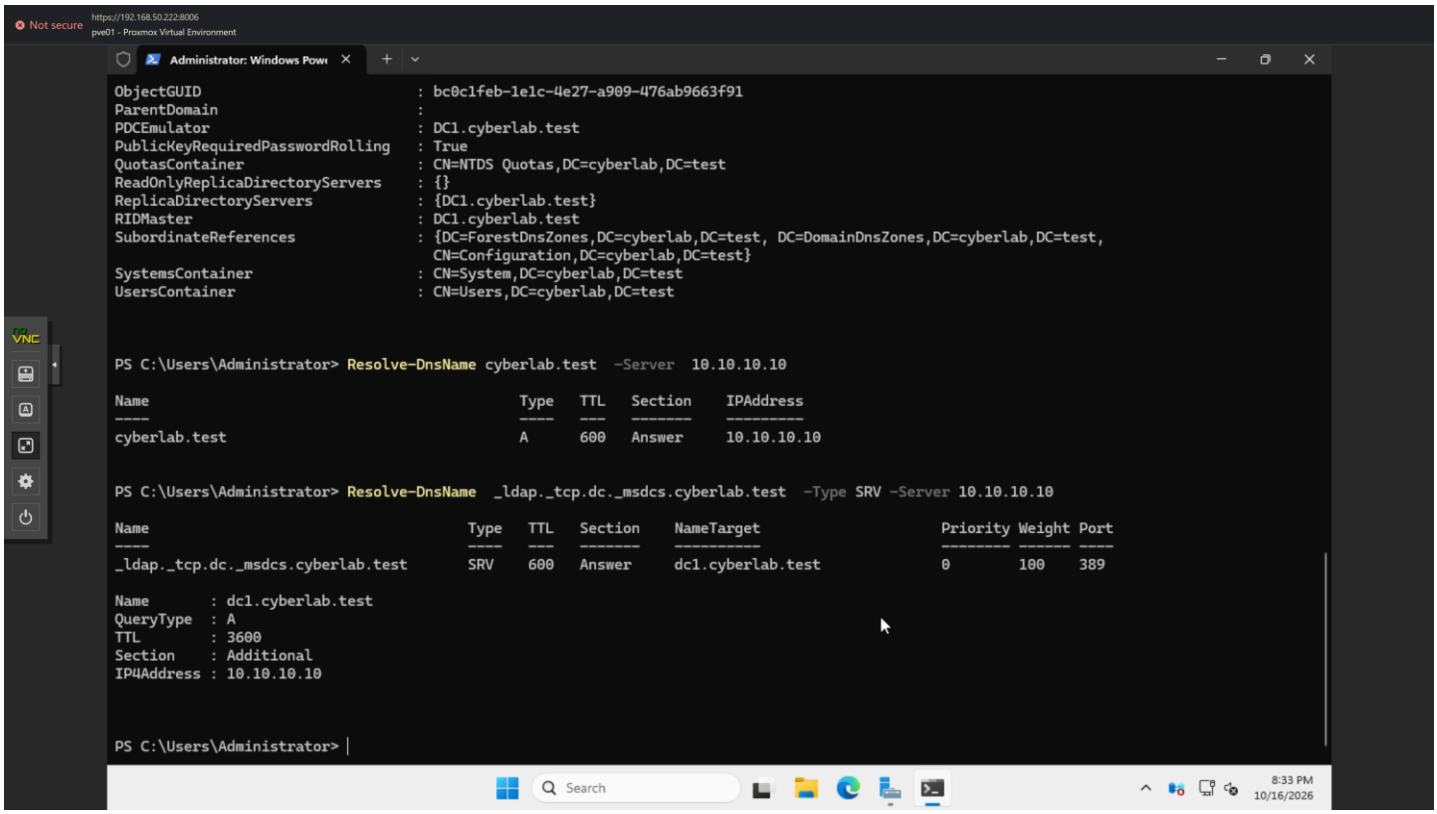


Figure 6. DNS returns cyberlab.test and the LDAP SRV record for dc1.cyberlab.test at 10.10.10.10.

DNS is part of domain authentication: the client needs the domain's service records to discover its controller. Working internet DNS alone does not establish that Active Directory discovery will work.

Directory organization and password controls

I created the CyberLab OU and child OUs for Users, Computers, Groups, and ServiceAccounts. I then created global security groups GG_IT_Admins, GG_Security_Analysts, and GG_Standard_Users. The student01 account became the standard-user test identity.

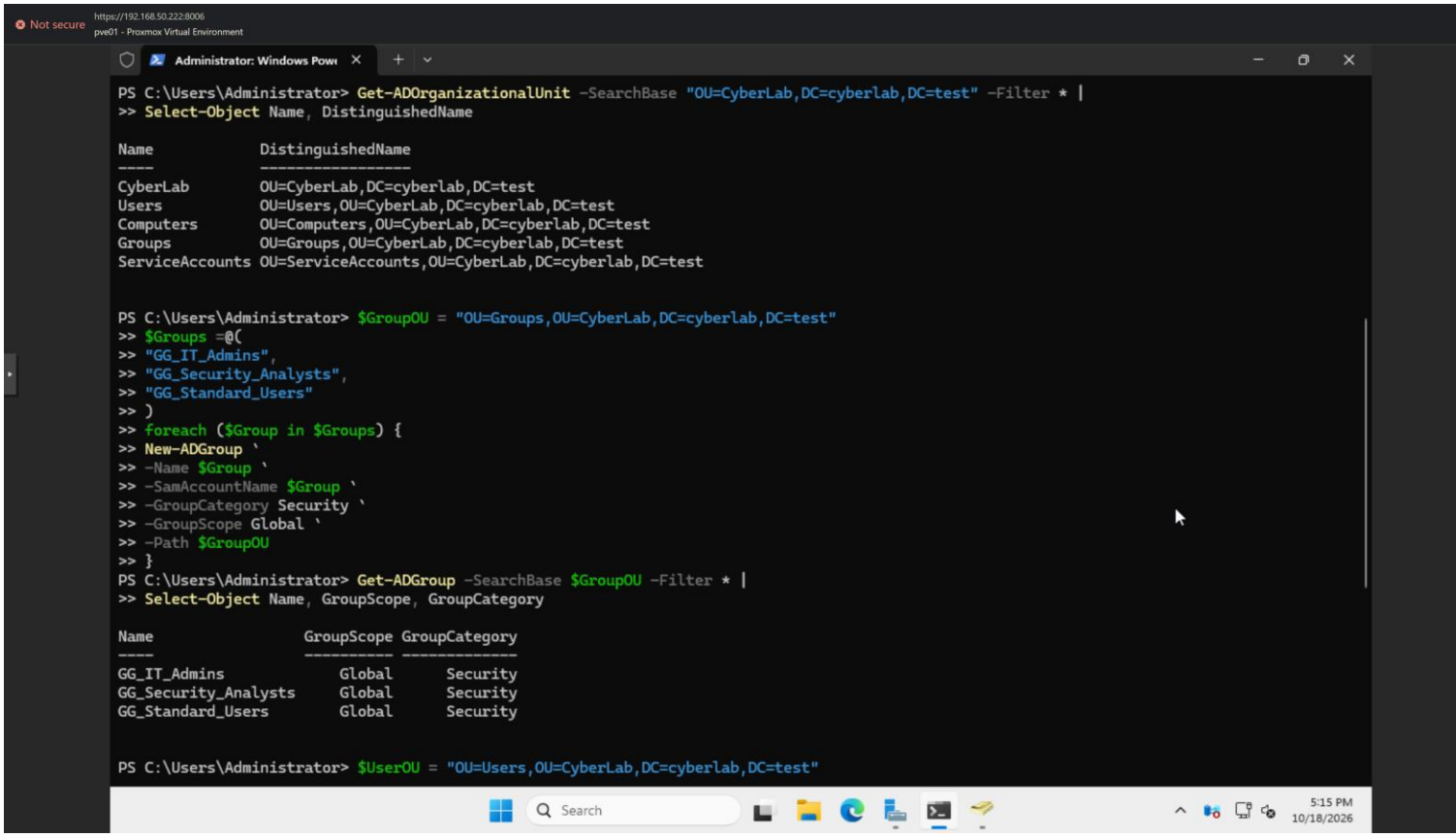


Figure 7. PowerShell verifies the OU structure and the three global security groups.

A group name does not grant privileges by itself. Group membership and the permissions assigned to that group must be verified separately. The later client checks demonstrate student01's standard-user context.

Policy setting	Earlier value	Later verified value
Minimum password length	7	14
Complexity	Enabled	Enabled
Password history	24	24

Policy setting	Earlier value	Later verified value
Lockout threshold	0	10 failures
Lockout duration and observation	10 minutes	15 minutes

The later policy screenshot is October 18 at 20:48:48. A 42-day maximum password age appears in the earlier policy output. The report records these lab settings as implemented values, not as a universal production password-policy recommendation.

Windows client networking and domain enrollment

I installed the Windows client and identified it as WIN11-CLIENT01. It obtained 10.10.10.108 on the lab subnet. I corrected the client DNS setting to 10.10.10.10 so domain discovery used DC1.

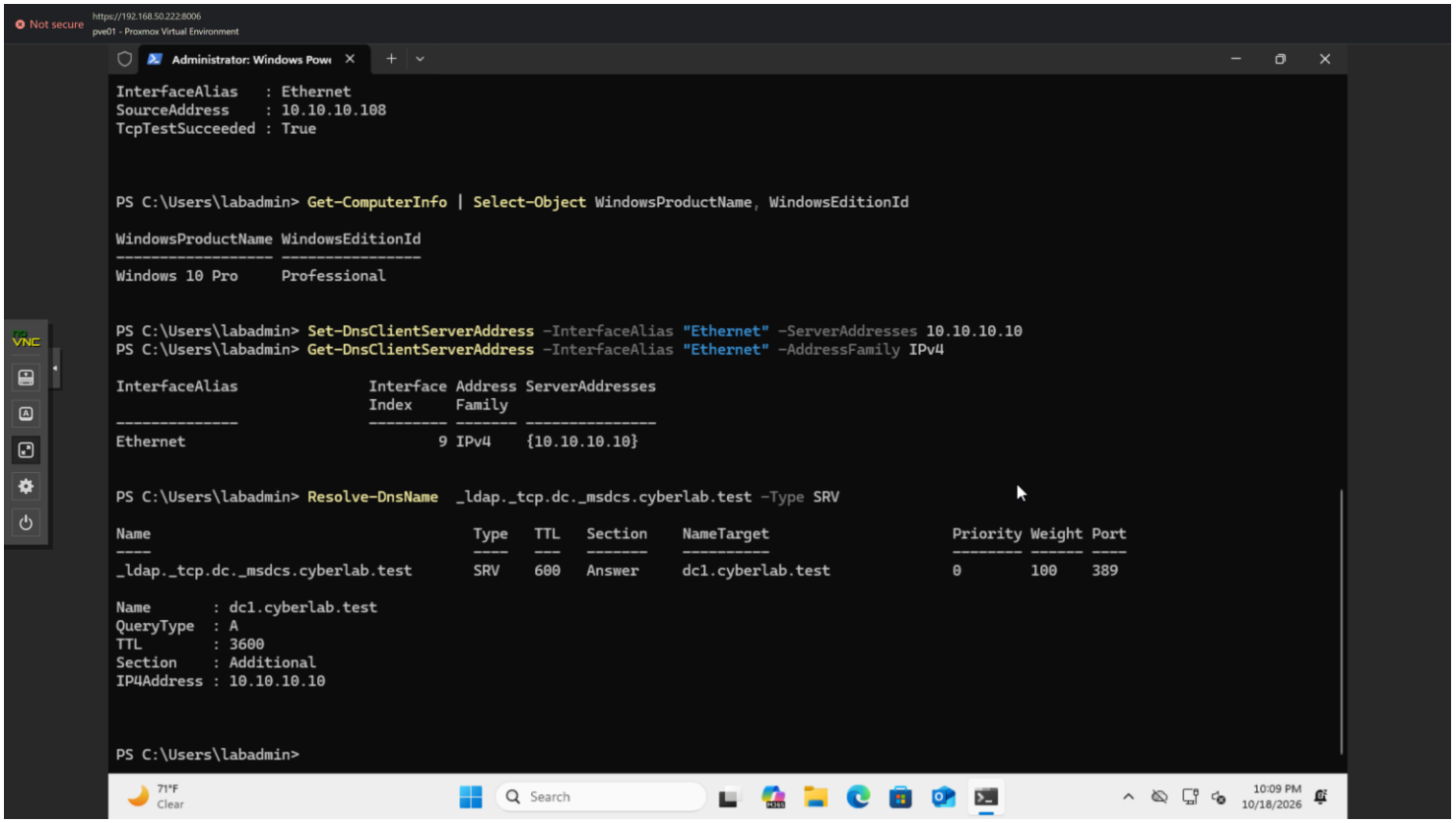


Figure 8. Client DNS is set to DC1, and LDAP SRV lookup resolves the domain controller.

The screenshot shows the Professional edition identifier. The WindowsProductName output reads Windows 10 Pro while the lab identifies the client as Windows 11; the edition and the successful domain workflow are more useful here than relying on that single product-name string.

Commands used in the enrollment sequence

The DNS command changes only the named adapter. The domain-join command prompts for an authorized lab credential and restarts the client after joining.

```
Set-DnsClientServerAddresses -InterfaceAlias "Ethernet" `
  -ServerAddresses 10.10.10.10

Add-Computer -DomainName "cyberlab.test" `
  -Credential "CYBERLAB\Administrator" -Restart
```

These commands document the build sequence; do not rerun a domain join on an already joined client. First inspect its current domain state. The restart interrupts the client session.

Successful domain sign in and least privilege

At first domain sign-in, student01 was required to change its password. An attempted replacement failed the configured policy. After using a compliant password, the client session reported cyberlab\student01.

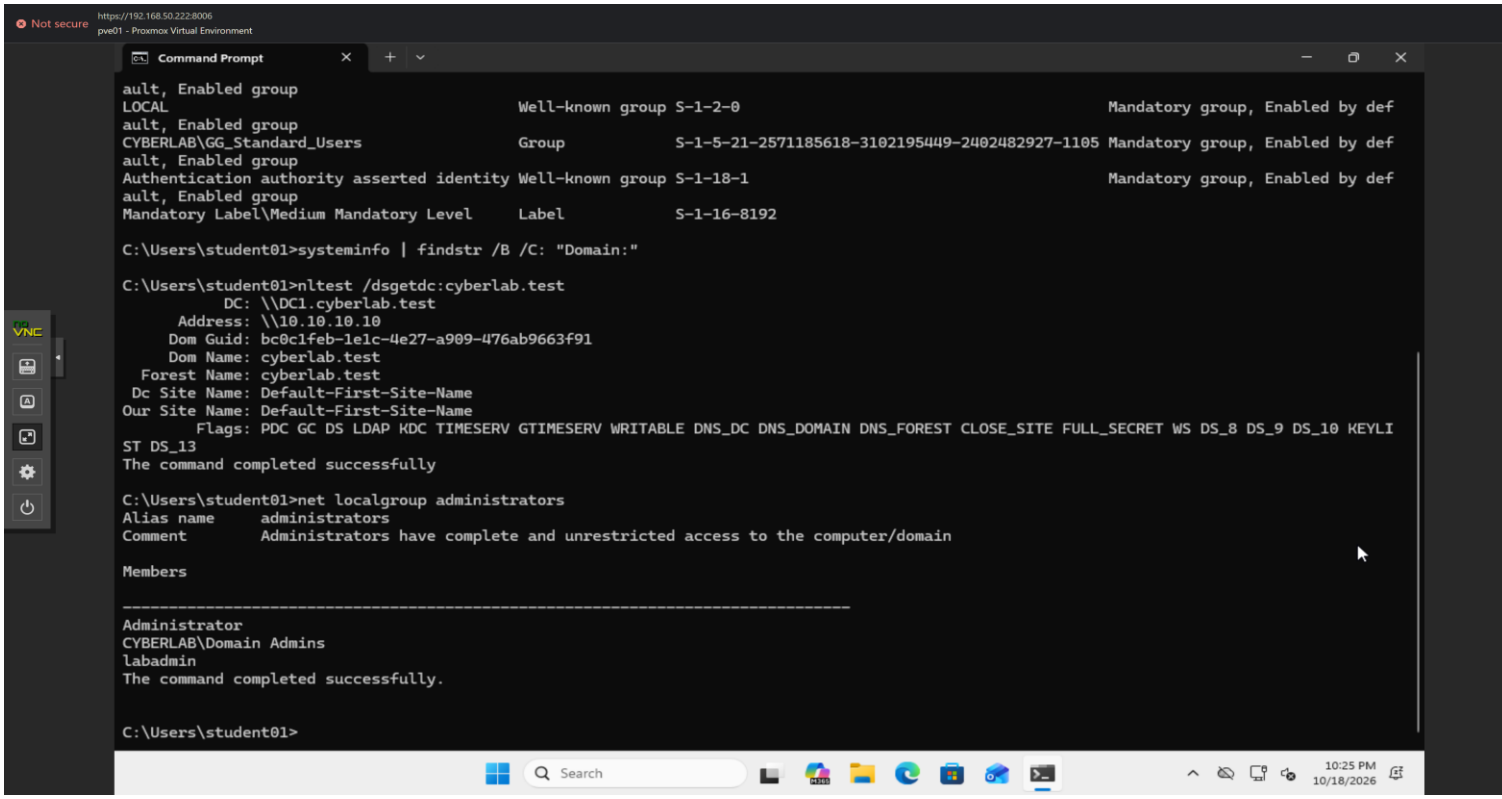


Figure 9. Client evidence shows GG_Standard_Users, medium integrity, successful DC discovery, and the local Administrators list.

Check	Observed meaning

Check	Observed meaning
GG_Standard_Users	The domain security group is present in the user token
Medium integrity	The observed session is not elevated
nltest /dsgetdc:cyberlab.test	DC1 is discovered at 10.10.10.10
Local Administrators	Administrator, CYBERLAB\Domain Admins, and labadmin are listed; student01 is not directly listed

Taken together, these checks support standard-user access for student01. Medium integrity alone would not rule out an administrator using a non-elevated token; the group and administrator-membership checks provide the necessary context.

DevSecOps relevance: separate everyday development identities from infrastructure administration and service identities. Test what an identity can actually do instead of relying only on its name.

Final setup post through Step 4

Steps 1–3 created Workstations under CyberLab, verified the OU, and located the client account. The initial lookup used WIN11-CLIENT and failed. Using the exact name WIN11-CLIENT01 returned its current object in the default Computers container.

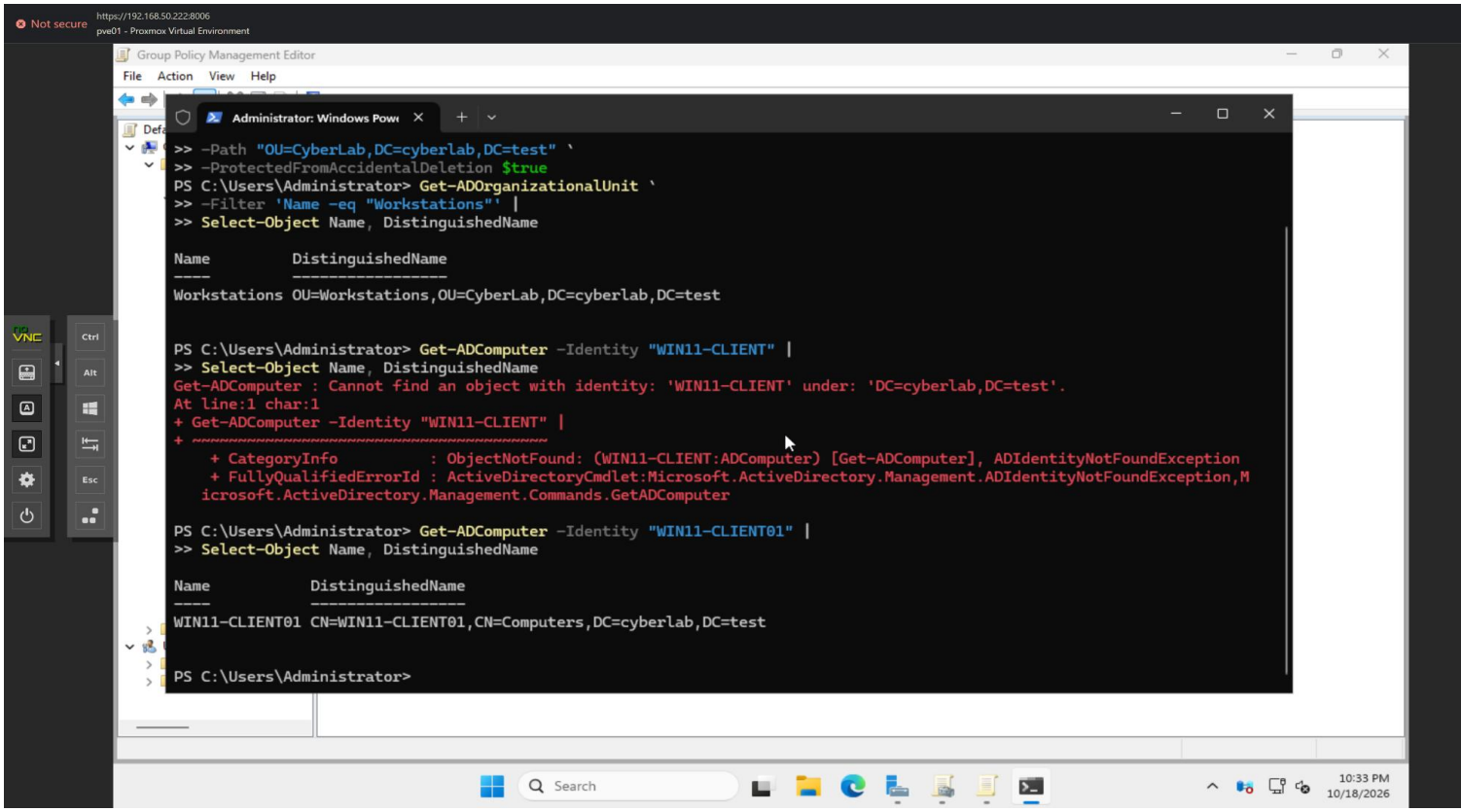


Figure 10. Workstations OU exists; corrected lookup locates WIN11-CLIENT01 in CN=Computers. Source: Continue Lab Configuration, final attached evidence.

Step 4 Move the client into Workstations

On DC1, in an elevated PowerShell session, the documented next change moves the computer object into the dedicated workstation OU. It does not change the VM's IP address. OU placement can change effective Group Policy, so inspect existing inheritance before applying the move.

```
Get-ADComputer -Identity "WIN11-CLIENT01" |
 Move-ADObject -TargetPath `
 "OU=Workstations,OU=CyberLab,DC=cyberlab,DC=test"
```

Status at the reporting boundary: prepared change, awaiting post-move verification. The latest captured distinguished name still shows CN=Computers. GPO creation, linking, and enforcement belong to a subsequent lab.

6 Verification Commands

Run these read-only checks on the stated machine to reproduce the functional baseline. The expected results describe acceptance criteria; they do not claim a new live test was performed for this report.

On the Windows client

Check identity, group membership, controller discovery, and local administrative membership.

```
whoami
whoami /groups
nltest /dsgetdc:cyberlab.test
net localgroup administrators
```

In PowerShell, inspect domain membership and DNS. Resolve the directory service record, then test the controller's LDAP and Kerberos TCP listeners.

```
Get-CimInstance Win32_ComputerSystem |
  Select-Object Name,Domain,PartOfDomain
Get-DnsClientServerAddress -AddressFamily IPv4
Resolve-DnsName _ldap._tcp.dc._msdcs.cyberlab.test -Type SRV
Test-NetConnection 10.10.10.10 -Port 389
Test-NetConnection 10.10.10.10 -Port 88
```

On DC1

Inspect the current computer location and domain password policy. The location check is the acceptance test after Step 4, not an assertion that the move has already succeeded.

```
Get-ADComputer -Identity "WIN11-CLIENT01" |
  Select-Object Name,DistinguishedName
Get-ADDefaultDomainPasswordPolicy |
  Format-List MinPasswordLength,ComplexityEnabled,`
  PasswordHistoryCount,LockoutThreshold,LockoutDuration
```

On Ubuntu and Proxmox

Ubuntu checks identify the host, addresses, route, and service states. On Proxmox, qm list inventories VMs. Keep host commands separate from guest commands.

```
# Ubuntu
hostname
ip -br addr
ip route
systemctl is-active ssh qemu-guest-agent
```

```
# Proxmox host
qm list
```

7 What Good Results Look Like

Area	Acceptance result
Lab addressing	Client is on 10.10.10.0/24 with the intended lab gateway
Directory DNS	Client DNS uses 10.10.10.10; LDAP SRV resolves to DC1
Domain membership	cyberlab.test and PartOfDomain=True
Identity	whoami returns cyberlab\student01
Least privilege	Expected standard-user group; no unintended administrative membership
Network isolation	Intended traffic works; prohibited traffic matches logged block rules
Step 4 acceptance	Computer DN includes OU=Workstations,OU=CyberLab
Reproducibility	Saved commands, configuration values, evidence, and recovery plan

8 Common Errors and Fixes

Observed problem	Resolution and lesson
SSH syntax error	Use username@host, such as root@192.168.50.222. Port 8006 is the web UI, not the SSH destination syntax.
Host-to-lab timeout	Identify the source network and firewall path; use VM console recovery before changing isolation.
Wrong computer name	WIN11-CLIENT failed; WIN11-CLIENT01 returned the correct directory object.
Unknown policy cmdlet	Get-ADDefaultDomainPolicy failed. The valid lookup was Get-ADDefaultDomainPasswordPolicy.

Observed problem	Resolution and lesson
Password change rejected	Choose a unique password meeting the configured policy; retain the security control.
DNS discovery problem	Point the lab client at DC1 and verify directory SRV records.
DHCP service uncertainty	Kea disabled and Dnsmasq enabled do not alone identify the active lease range or DNS options.
Linux command in Windows	Use ping -n 4 on Windows; ping -c 4 is the Linux count syntax.

Findings and Remediation

Risk levels below describe the potential consequence in this lab, not a formal production assessment. Open items should be resolved before adding vulnerable workloads or unattended deployment jobs.

Finding	Risk	Remediation and closure evidence
IPv4 home-network blocking demonstrated	Reduced	Retain rule export and log evidence; retest after network changes.
IPv6 and alternate paths need validation	High if reachable	Test IPv6 policy and adapter/bridge paths; document both allowed and denied traffic.
Lab workloads share a subnet	Medium	Separate intentionally vulnerable targets from identity and management systems; verify the resulting boundary.
Client DNS manually corrected	Medium	Verify active DHCP scope and option 6 before distributing DC1 DNS automatically.

Finding	Risk	Remediation and closure evidence
Step 4 move not yet evidenced	Low now	Capture the post-move distinguished name before linking workstation policy.
Dedicated workstation baseline pending	Medium	Build a scoped GPO, pilot it, and verify effective settings and logs in the next lab.
Backup and restore proof needed	Medium	Export firewall configuration and back up VMs; perform a documented restore exercise.

9 What to Screenshot for Documentation

Keep the original captures and use one clear image per claim: topology and adapter mapping; firewall rule order plus matching log; DC1 addressing and DNS records; OU/group structure; policy values; client DNS; successful domain identity and group checks; and the final computer distinguished name after Step 4.

Each new capture should record the machine, command or UI view, date, expected result, and observed result. Preserve evidence of failures when it explains the fix. Remove credentials, tokens, and unrelated personal account details before sharing a portfolio copy.

10 Reflection Questions

1. Why could a client have internet access but still fail to find its Active Directory domain?

2. What makes a firewall log entry stronger evidence than a failed ping alone?

3. Why does moving a computer into an OU potentially affect security without changing its IP address?

4. Which checks distinguish a standard user from an administrator running a non-elevated session?

5. Which manual steps can become repeatable scripts, and what test proves each script worked?

6. What recovery evidence would let me rebuild this lab after a failed change?

Lessons learned

I made the lab functional by checking dependencies in order: virtual machines and interfaces, addressing and routing, DNS, directory services, then client authentication. Exact hostnames and cmdlet names mattered. Failed commands were useful when I compared them with corrected commands and their output.

The most important change toward DevSecOps is to treat security controls as testable engineering requirements. Configuration alone is incomplete; I need reproducible setup, a validation result, useful logs, and a recovery path.

11 Next Lab Recommendation

Beginner next lab: verify Step 4, then create and test a dedicated workstation security baseline. Keep it scoped to Workstations. Validate firewall, endpoint protection, auditing, and logging with effective-policy evidence before expanding the scope.

Intermediate next lab: build a Git-managed Ubuntu baseline with SSH key access, a Bash or Python validation script, and an Ansible configuration. Demonstrate a clean rebuild and an idempotent second run.

Advanced next lab: build a small application, containerize it, and add CI checks for tests, secrets, dependencies, and source code. Deploy only into a separate development segment. Preserve scan findings, fixes, release artifacts, and rollback evidence.

DevSecOps Portfolio Direction

Period and focus	Concrete project outcome	Evidence to retain
Now through January 2027 **Stable foundation**	Verify network boundaries, finish workstation organization, practice Linux, and prepare for Security+	Network diagram, configuration exports, least-privilege checks, restore record
2027 **Code and automation**	Learn Python, Bash, Git, and configuration automation alongside coursework	Readable scripts, version history, automated tests, rebuild instructions
2028 **Secure delivery**	Add Docker, CI/CD, and a small cloud/IaC learning project	Pipeline runs, scan reports, remediations, IAM design, teardown procedure

Period and focus	Concrete project outcome	Evidence to retain
2029 and onward Integrated capstone	Demonstrate application delivery, security testing, monitoring, and response	Threat model, release evidence, incident exercise, technical portfolio narrative

These milestones follow the shared career discussion through its final section 4 on home-lab priorities. They are a proposed learning sequence. Docker, cloud infrastructure, Kubernetes, and CI security tools are not claimed as deployed components of the present lab.

How the present work transfers

Proxmox and VM troubleshooting develop infrastructure operations skills. OPNsense rules and logs develop segmentation and observability. Active Directory develops identity and least privilege skills. PowerShell and Linux checks provide a starting point for automated compliance tests. The next bridge is software development: creating an application whose build, test, deployment, and security results can be demonstrated end to end.

Portfolio statement

Built and documented a Proxmox-based cybersecurity lab with OPNsense filtering, Active Directory DNS, and a domain-joined Windows client. Validated domain discovery, standard-user identity, and logged IPv4 lab-to-home blocking; prepared a dedicated workstation OU for the next security-baseline stage.

Evidence Sources

Primary build evidence: Manniesoma Workspace screenshots dated October 16 and October 18, 2026, plus the 71-page Binder1.pdf. The binder preserves the earlier provisioning sequence; selected original screenshots support the later operational milestones.

Conversation sources: Continue Lab Configuration (final setup sequence and OU evidence); SSH connection steps (remote-access syntax and inventory); Career Relevance Analysis (academic context and DevSecOps direction).

Figure source index

1. Binder1.pdf, page 25
2. Binder1.pdf, page 37
3. Screenshot 2026-10-16 174021.png
4. Screenshot 2026-10-16 194839.png

5. Screenshot 2026-10-16 200705.png

6. Screenshot 2026-10-16 203337.png

7. Screenshot 2026-10-18 171528.png

8. Screenshot 2026-10-18 220904.png

9. Screenshot 2026-10-18 222543.png

10. Continue Lab Configuration, final attachment 4c949de6-b04e-4cf4-92e2-9f2b69ff05de.png

Additional corroborating captures

Screenshot 2026-10-18 203237.png records the earlier password-policy values; Screenshot 2026-10-18 204848.png records the later values and corrected policy cmdlet. Screenshots 2026-10-18 220923.png and 220929.png record Dnsmasq settings. Screenshot 2026-10-18 222233.png records the successful domain identity.

Historical addresses, software-version labels, and service values are recorded as shown at capture time. They should be rechecked before using this report as an operational runbook.